AYE
TECH HUB

• HVAC ENGINEERING • LESSON 07

Duct Design

Basics

Lesson 07 of 30 | HVAC Engineering Program

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- Rectangular vs circular ducts — when to use each
- The equal friction method — sizing ducts step by step
- Velocity limits by space type — noise control in design
- Duct fitting pressure losses: elbows, tees, reducers and dampers
- Reading and producing a duct sizing schedule like a professional

Duct Design — Moving Conditioned Air to Every Zone

A duct system is the air delivery network that connects the AHU (Air Handling Unit) to every supply and return grille in the building. **Good duct design delivers the right volume of air to every zone at the right velocity, without excessive noise or energy waste.** Bad duct design — oversized ducts, poor routing, inadequate sealing — is one of the most common causes of HVAC system underperformance.

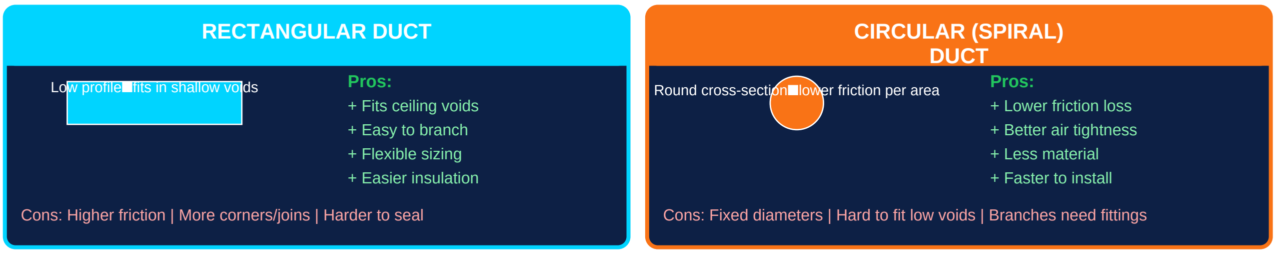

Figure 1 — Rectangular vs Circular Duct Comparison

Duct Material	Typical Use	Advantages	Limitations
Galvanised steel (rectangular)	Main trunk ducts, plant rooms, commercial buildings	Strong, durable, good acoustic properties, well-established detailing	Higher cost, heavier, more joints to seal, skilled fabrication needed
Spiral steel (circular)	Branch ducts, exposed installations, clean rooms	Lower pressure drop, fewer leaks, faster installation, cleaner look	Fixed diameters, harder to route in tight spaces, larger bend radius
Flexible duct (insulated)	Last metre connection to grilles only — never for main runs	Absorbs vibration, easy to connect grilles, no fabrication	High friction loss — max 1.5m length. Never bend sharply. Collapses if squashed.
GRG / Phenolic foam duct	Cleanrooms, food factories, low-temperature supply	Smooth bore, excellent air tightness, thermally insulated in one product	Brittle, requires specialist installation, higher material cost

Duct leakage: UK CIBSE and EU standards define duct leakage classes (Class A lowest to Class D lowest). A Class C system leaks ≤ 0.25% of design flow per metre of duct. In practice, poorly sealed duct systems can lose 15–25% of supply air through leaks — wasting energy and failing to deliver air where it is needed.

The Equal Friction Method — Sizing Ducts Step by Step

The **equal friction method** is the most widely used duct sizing technique. The principle: design all ducts so that the pressure loss per unit length is equal throughout the system — typically 0.8–1.2 Pa/m. This ensures that each branch of the system is inherently self-balancing, reducing the need for dampers.

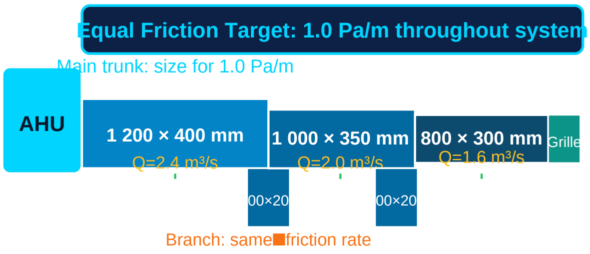

Figure 2 — Equal Friction Method: Duct Reduces as Flow is Distributed

Equal Friction Method — Step by Step

Step	Action	Example
1—Define supply air quantities	Calculate airflow for each room from ventilation or cooling load. Sum all flows to get total AHU supply air.	Room 1: 0.4 m^3/s, Room 2: 0.4 m^3/s, Room 3: 0.4 m^3/s × 6 rooms = 2.4 m^3/s total
2—Choose friction rate	Select target friction rate. Use 1.0 Pa/m for typical commercial, 0.8 Pa/m for quiet offices, 1.5 Pa/m for plant rooms.	1.0 Pa/m target for a commercial office building
3—Size main trunk duct	Use duct sizing chart or software. Find size for 2.4 m^3/s at 1.0 Pa/m. Check velocity is within limits for the space.	1200 × 400 mm at 5.8 m/s — acceptable for plant room, acceptable velocity
4—Size each branch after take-off	After each branch takes flow, resize the main duct for the remaining flow. Size branch duct for its flow at the same 1.0 Pa/m.	After branch 1 (0.4 m^3/s), main duct carries 2.0 m^3/s → resize to 1000×350mm

5—Calculate total pressure loss	For the index circuit (longest/highest resistance path): sum all friction losses + fitting losses.	Index circuit total: 8+6+4 m × 1.0 Pa/m + fittings = 28–40 Pa total
6—Select AHU fan	Fan must provide design flow rate at design external static pressure. Add 20% safety margin for unforeseen losses.	Fan duty: 2.4 m^3/s at 400 Pa external static (total system + terminal units)

Maximum Design Velocities by Space Type

Space Type	Main Duct (m/s)	Branch Duct (m/s)	Grille Face Velocity (m/s)	Noise Concern
Plant room / basement	8–12	6–8	—	None — not occupied
Office (open plan)	5–7	4–5	2.0–2.5	Low — background noise acceptable
Quiet office / meeting	4–6	3–4	1.5–2.0	Medium — conversation must not be disturbed
Library / courtroom	3–4	2.5–3	1.0–1.5	High — near-silence required
Hospital ward / bedroom	3–4	2.5–3	1.0–1.5	High — patient comfort and sleep
Kitchen extract	8–12	6–8	—	High grease load — clean regularly

Duct Fittings, Losses and the Duct Schedule

Straight duct sections cause friction loss proportional to length. But fittings — elbows, tees, reducers, grilles — cause additional **minor losses** that can be significant. Professional duct design accounts for all fitting losses in the pressure drop calculation.

Fitting Type	Zeta Value (ζ)	Pressure Loss Formula	Note
90° mitre elbow (no vanes)	1.2–1.5	$\Delta P = \zeta \times (\rho v^2/2)$	Avoid — very high loss. Use radiused elbows or turning vanes instead.
90° radiused elbow (R/W = 1.5)	0.2–0.4	$\Delta P = \zeta \times (\rho v^2/2)$	Standard for most installations. Specify radius on drawings.
45° elbow	0.1–0.2	$\Delta P = \zeta \times (\rho v^2/2)$	Lower loss than 90°. Use where routing allows.
Tee — main flow	0.05–0.10	Based on main velocity	Straight through a tee has low loss
Tee — branch off	0.5–1.0	Based on branch velocity	Velocity change at branch causes loss
Abrupt contraction (reducer)	0.3–0.5	Based on downstream velocity	Taper over 15° max to minimise loss
Supply grille / diffuser	1.0–3.0	Based on neck velocity	Select grille so neck velocity ≤ 3 m/s
Volume control damper	0.2 (open) to ∞ (closed)	—	Minimum pressure loss when fully open. Use for balancing, not throttling.

The Duct Sizing Schedule

DUCT SIZING SCHEDULE — AHU-01 Supply Air System							
Duct Ref	Flow (m^3/s)	Velocity (m/s)	Size (mm×mm)	Equiv Dia (mm)	Fric Loss (Pa/m)	Length (m)	Total ΔP (Pa)
M-01	2.40	5.8	1200×400	780	1.02	8.0	8.2
M-02	2.00	5.5	1000×350	672	0.98	6.0	5.9
M-03	1.60	5.2	800×300	550	1.05	4.0	4.2
B-01	0.40	3.5	400×200	278	0.95	3.0	2.9
B-02	0.40	3.5	400×200	278	0.95	4.0	3.8

Figure 3 — Sample Duct Sizing Schedule

Schedule Column	What It Records	Where It Comes From
Duct Reference	Unique ID for each duct section (M-01, B-01, etc.)	Assigned on the duct layout drawing
Flow rate (m^3/s)	Airflow in that duct section	Calculated from room loads or ventilation schedule
Velocity (m/s)	Air speed in duct = Q / A	Calculated from duct size — must be within limits for space type
Duct size (mm×mm)	Width × Height for rectangular; dia for circular	From duct sizing chart at target friction rate
Equivalent diameter	De = 1.265 (ab/(a+b))^0.5 for rectangular	Used to read circular duct pressure drop from chart
Friction loss (Pa/m)	Pressure drop per metre — should equal target (e.g. 1.0 Pa/m)	Read from Moody chart or duct sizing software

Section length (m)	Length of this duct section on drawing	Scaled from duct layout drawing with allowance for fittings
Total ∆P (Pa)	Length × Pa/m + fitting losses	Summed for index circuit to determine fan duty

What comes next — HVAC-008: Air Distribution Systems — supply and return grille types (swirl diffuser, linear slot, perforated tile), throw distance, Coanda effect, displacement ventilation, and how to layout diffusers for even temperature distribution.